Vortex86DX3

(A9126) Layout Guideline

Ver. 0.8

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Modify	Date (mm/dd/yy)	Version	Description
Mango	06/17/13	0.1	Initial Release
Mango	04/08/14	0.2	1. Add PCIe external CLK guideline 2. Add USB、PLL、PCIe、SATA、Ethernet、DVO power layout guideline 3. Add Ethernet、USB ESD layout guideline 4. Add PCIe reset signal recommend circuit 5. Add ADC protection circuit
Mango	04/18/14	0.3	1. Modify Ethernet protection scheme for connect to POE device
Mango	07/03/14	0.4	1. Add VID layout guideline
Mango	03/13/15	0.5	1.Add VGA layout guideline
Mango	04/10/15	0.6	1, Modify VGA layout guideline
Sanders	05/14/15	0.7	1. Update the schematic of VGA part to DM226B
Sanders	01/25/17	0.8	1, Modify the 1.16 Power on sequence (remove Without RTC Battery)

Overview

1.1. General Description

The following document provides some technical data that describe layout guideline of VORTEX86DX3. The main purpose of this document was to provide VORTEX86DX3 with high quality signal for system reliable operation.

1.2. IDE bus Layout Guideline

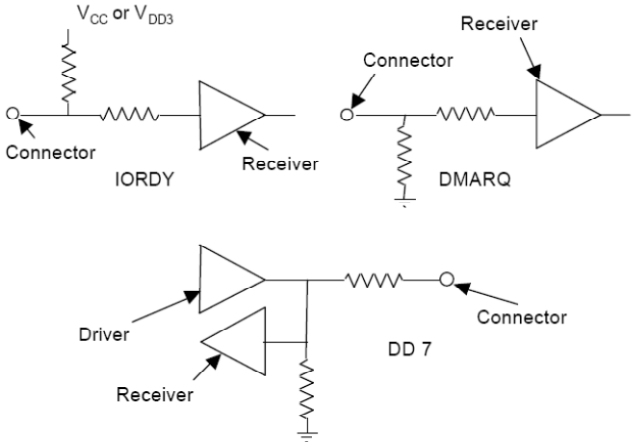
Matches trace length of IOR- and IORDY within 100mil. Matches trace length of IOR-/IORDY and D0~D15 within 400mil. Control the trace length of IDE interface within 6 inch. Routing signals of IDE with 5mil trace width and keep minimum 10mil space. We recommend to buffer DASP- signal to driver LED to reduce EMI issue. The pull up and pull down is required for interface on host side as follows:

1. The DMARQ needs a 5.6k ohm pull down resistor.
2. The IORDY needs a 4.7k ohm pull up resistor.
3. The INTRQ needs a 10k ohm pull down resistor.
4. The CSEL connect to ground.
5. DASP- needs a 100k ohm pull up resistor if buffer the signal to driver LED.

The series termination required for the interface on host side as follows:

1. The IOR-, IOW- and DMACK- need a 22ohm series termination resistor.
2. The CS0-, CS1-, DA0~2, D0~15 and RESET- need a 33ohm series termination resistor.
3. The DMARQ, INTRQ and IORDY need an 82ohm series termination resistor.

The topology for series termination with pull up or pull down is as follows:



Place all series termination near IDE connector.

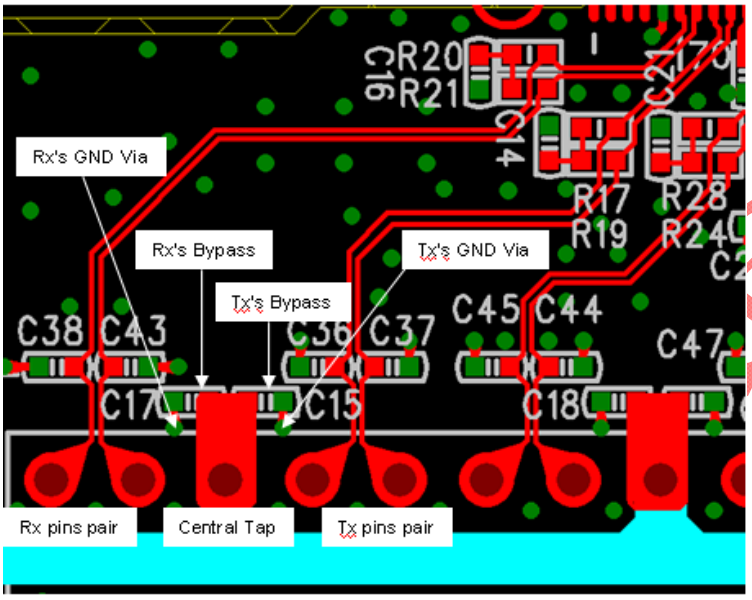
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1.3. Ethernet PHY Layout Guideline

Place termination resistor near IC side. Route differential pair of Ethernet PHY with 4.5mil traces width and 5mil space between TXP (RXP) and TXN (RXN), keep minimum 30 mils space for other traces to reduce crosstalk and EMI as possible. Keep the analog signals away from digital signals. Place resistor of ISET near VORTEX86DX3 at least with trace width of 20mil. Place Transform near RJ45 connector and place termination resistor near IC side.

Pin A11 (AVSS1_EPHYTX18) is the ground pin of transmitter; it can't share via with other analog ground pins. It can't through bead connect to system ground. Strong suggestion connection to clear system ground directly.

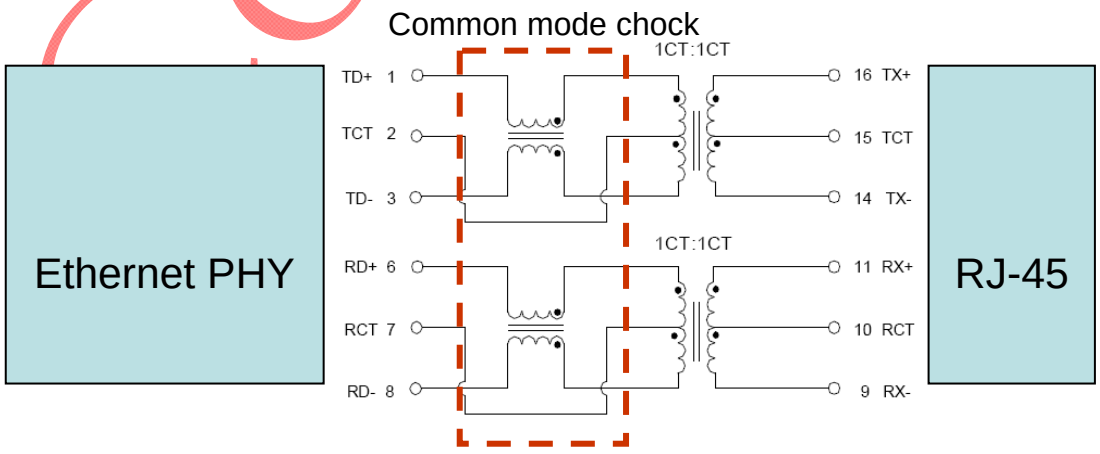
It is importance for the layout of transformer's central tap. Due to our PHY support auto MDI/MDIX. Add the bypass capacitor of central tap for transmit path and receiver path with ground via near transmit and receiver pins of transformer respectively. The figure is as follows:



Ethernet's Layout guideline

The LAN transformer selection guide is as follow:

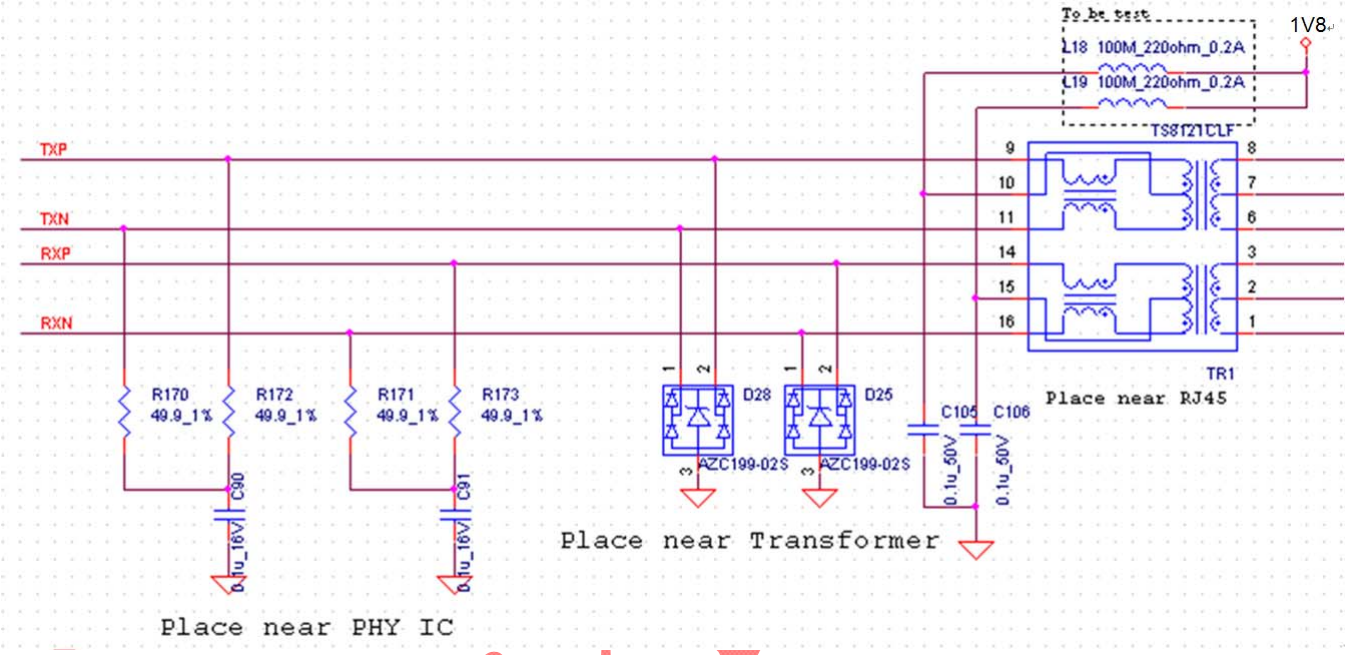
Select the LAN transformer with common chock on the PHY side to enhance EMI's performance.



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1.4. Ethernet protection scheme for connect to POE device

To avoid surge pulse cause Ethernet PHY damage, place TVS components to resist the surge pulse when connect with POE device, and TVS protect circuit have to use 2 high-speed I/O lines specification of TVS for each TXP/N and RXP/N differential pairs , can not use 4 high-speed I/O lines specification of TVS , both of TVS component place near by LAN Transformer as following:



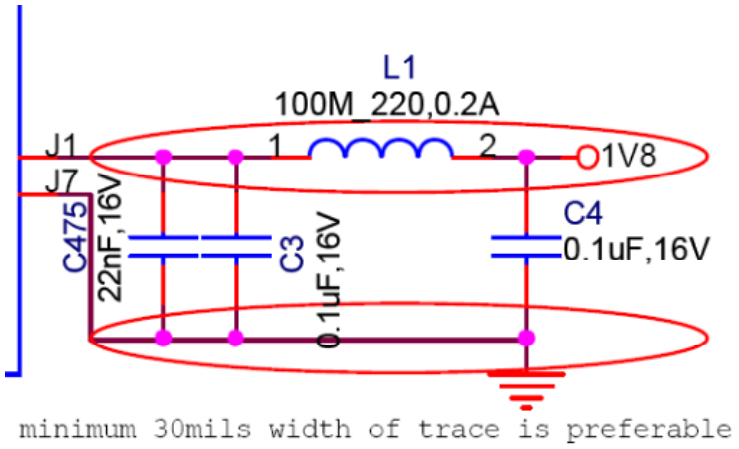
1.5. USB PHY Layout Guideline

Route differential pair of USB PHY with 5mil traces width and 5mil space between DP and DN, keep minimum 30 mils space for other traces to reduce crosstalk and EMI as possible. Keep the analog signals away from digital signals. Place the resistor of ISET near A9130 at least with trace width of 20mil. Place common chock near connector if present.

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1.6. PLL layout Guideline

A clear power supply is importance for PLL in high performance CPU. DMP recommend a filter technology to reduce jitter as follows:



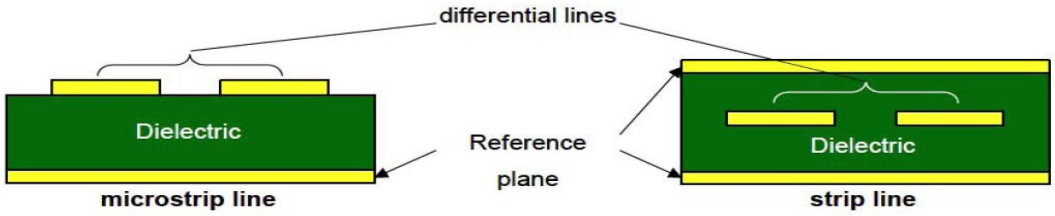
Rely on VORTEX86DX3 IC side the second layer must be the ground layer.

1.7. SD Card Interface Layout Guideline

Route traces length of SD Card Interface within 1000mil.

1.8. SATA and PCI EXPRESS Interface Layout Guideline

1. The characteristics impedance of transmission lines must be 100 ohm +/- 10% for differential mode and maximum 35 ohm for common mode of SATA and PCIe connection in principle. For appropriate characteristics impedance, the strip line or microstrip line structure must be used. Selection of strip line or microstrip line should be examined on a case-by-case basis according to package (socket), pin count and macro count.

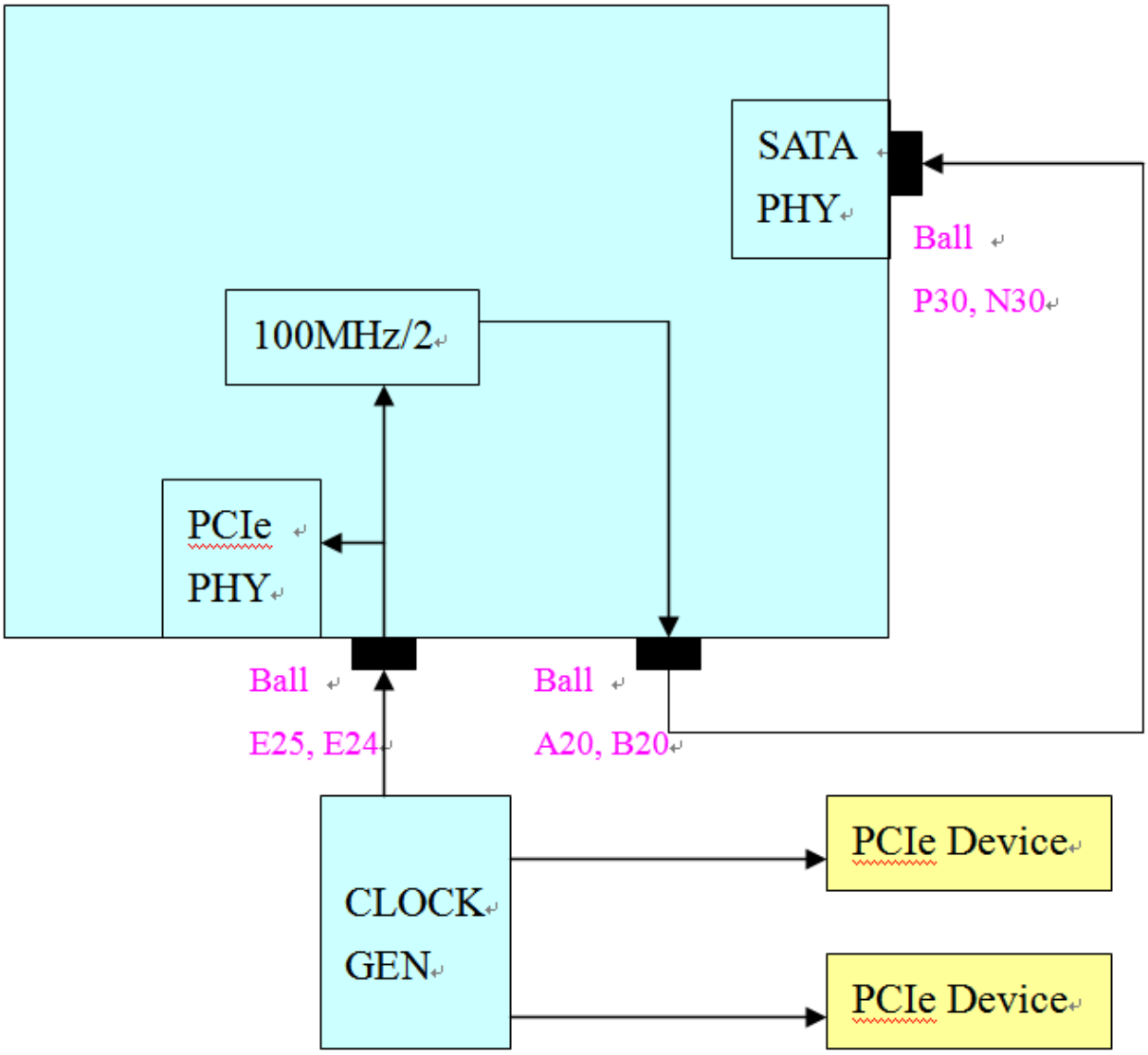


Differential lines structures

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2. SATA RX and SATA PLL can common power but, SATA TX power is absolutely not in front of two shared power.

1.9. SATA and PCIe CLOCK SOURCE



The PCIe clock source

1. **external clock GEN 100MHz**

The Pcie Device Clock source

1. **external clock GEN 100MHz**

The SATA clock source

1. **PCIe clock 100MHz/2(External Clock GEN)**

1.10. 1G bytes DDR3 Layout Guideline

According to the below-mentioned guideline, some constraints were appended as follows:

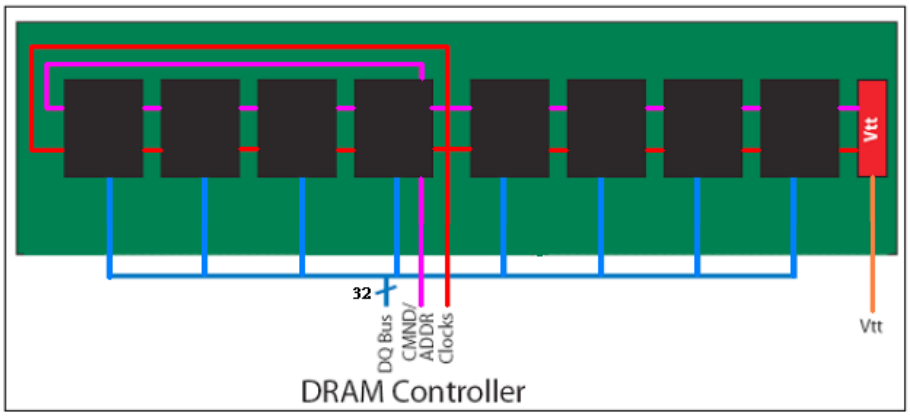
1. Control the impedance of trace between 60 and 65 ohm.
2. Control the trace length of DQS/DQS# between 1700 and 2000 mil.
3. Control the trace length of DQ/DQM is shorter than DQS/DQS# between 0 and 700 mil.
4. The damping resistor is 10 ohm for address, command and data bus.
5. Control the trace length of CLKP/CLKN is longer than DQS/DQS# between 0 and 300 mil

1.11. DDR3 Interface layout Guideline

More attention must be paid to high speed rule for board layout and bus loading to make sure that short clock cycle time can be met without noise, ringing, crosstalk power or ground bounce. To provide a good return path and limit above noise on the DDR interface, all system memory signals should reference a single contiguous plane as possible.

The recommendation layout guideline for a DDR device is as follows:

1. In order to attain highest speed as possible. We suggest the signals of DQS/DQS#, DQM and DQ only a loading preference. Please connect with DMP if the signals of DQS/DQS#, DQM and DQ need to load more than one.
2. DDR3 uses a “fly-by” topology in the module designs as seen in Figure 1. This daisy chained from one DRAM component to the next. The fly-by topology helps in achieving shorter stub lengths to the DRAM and reducing the number of stubs. Due to this the timing at each DRAM does not need to be matched for each net and simultaneous switching noise is reduced. The VTT termination off of the motherboard to the end of the Address/Command/Control bus. Larger bandwidth and improved SI are achieved when proper VTT termination is used in the fly-by topology. The fly-by topology does introduce interconnect delay skew between the clock and data strobe at each DRAM.

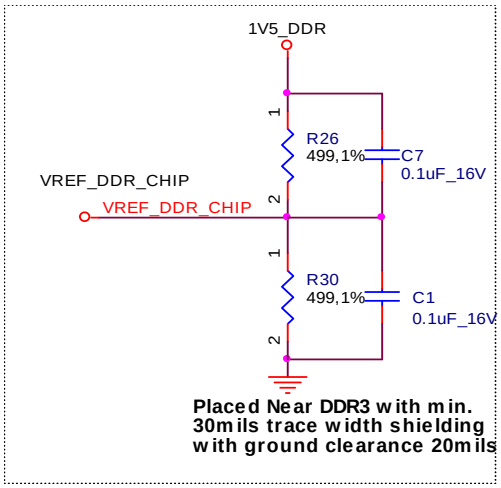


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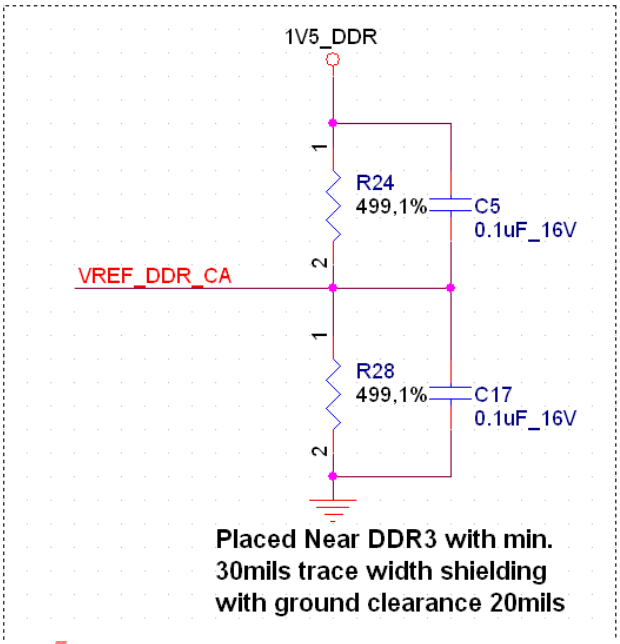
3. The DDR clock is differential signal that need a 100 ohm differential terminal resistor near DDR side for a DDR configuration. Reserve location of two 0ohm damping resistors near CPU side for low power application. The two trace width is 4.5 mil and space 5mil to get near 100ohm differential impedance, and keep minimum 15 mils space for other traces to reduce crosstalk as possible. The trace length of two differential signals must be match within 20mil. The category signals were routed within 4.6 inch.
4. Routing each signals of byte lane on same layer with same via count. Control the number of via within 2. The mismatch of trace length must be within 20mil for each signals of byte lane. The high byte lane is DQS1/DQS1#, DQM1 and DQ8~DQ9 and the low byte lane is DQS0/DQS0#, DQM0 and DQ0~DQ7 respective. Place a damping resistor on middle of each signal to trade off the timing of read and write for low power with ODT off. The DQS/DQS# is differential signal that they were routed differential with 4.5mil trace width and 5mil space between DQS and DQS#, keep minimum 15 mils space for other traces to reduce crosstalk as possible. The other signals were routed 5mil trace width and keep minimum 10mil space between signals to reduce crosstalk and control impedance of trace. The high byte lane and low byte lane must be matching within 0.2 inch. The category signals were routed within 1.2 inch. It is importance to match the differential signals of DQS and DQS# within 10mil.
5. Routing signal of command and address with a 0ohm damping resistor near CPU side. The category signals were routed 4~5mil trace width with minimum 10mil space between signals to reduce crosstalk and control impedance of trace. The signals of command and address must be matching within 0.4 inch. The category signals were routed within 5 inch.
6. To provide a good return path, all system memory signals should reference a single contiguous plane as possible.
7. DDR memory signal trace edges must be at least 4 times (20 mils) trace width from any non-DDR memory signal trace edge.
8. Routing trace near edge of the reference plane should keep minimum 5 times trace width from edge of signal trace to edge of the reference plane.
9. The signals of address and command must be following “Fly-by” routing rule for the application of two DDR devices.
10. The generation of Vref is critical. It must be follow the variance of VDDQ. We are suggestion to generate the reference from VDDQ by two 499ohm divide resistors with parallel 0.1uf X7R type bypass capacitor for each resistor. Place the divide resistors and bypass capacitor on quiet area. Also place a 0.1uf X7R type bypass capacitor on Vref pin of each device.

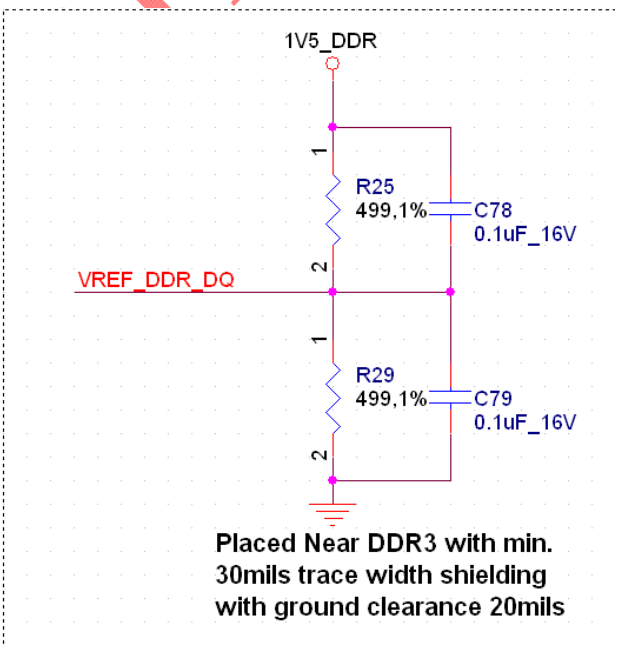
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11. The circuit of Vref is as follows:



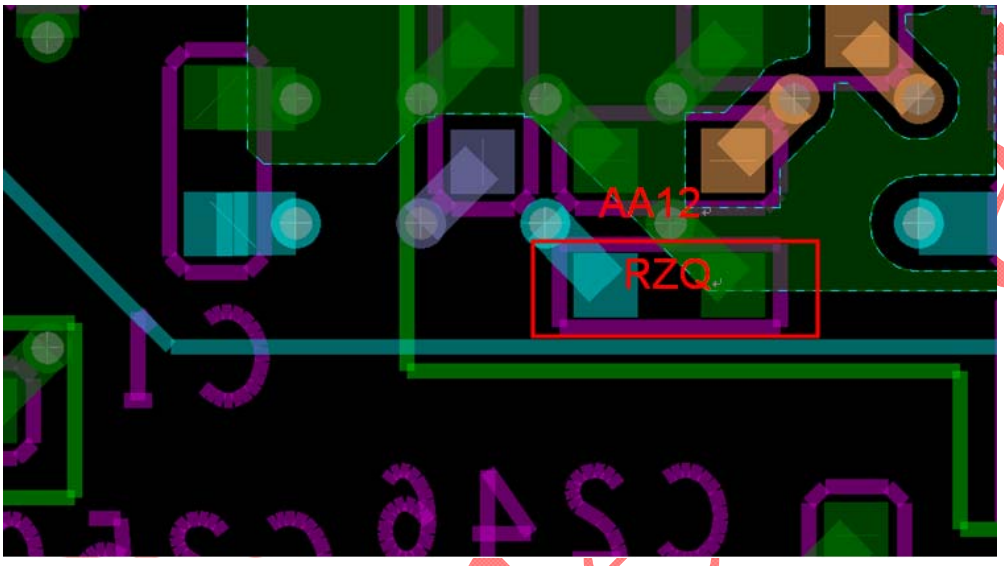
Vref power have two independent reference voltage for DDR3 device , VREF_DDR_DQ provide the reference voltage for DQ, VREF_DDR_CA provide the reference voltage for CA





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The DDR3 interface reference voltage of VORTEX86DX3 is critical for power noise. In order to reduce the power noise of reference voltage and follow the DDR3 interface voltage of VORTEX86DX3. Please place the RZQ under the VORTEX86DX3 next to the pin AA12. The layout of suggestion is as follows for the DDR3 reference of VORTEX86DX3:

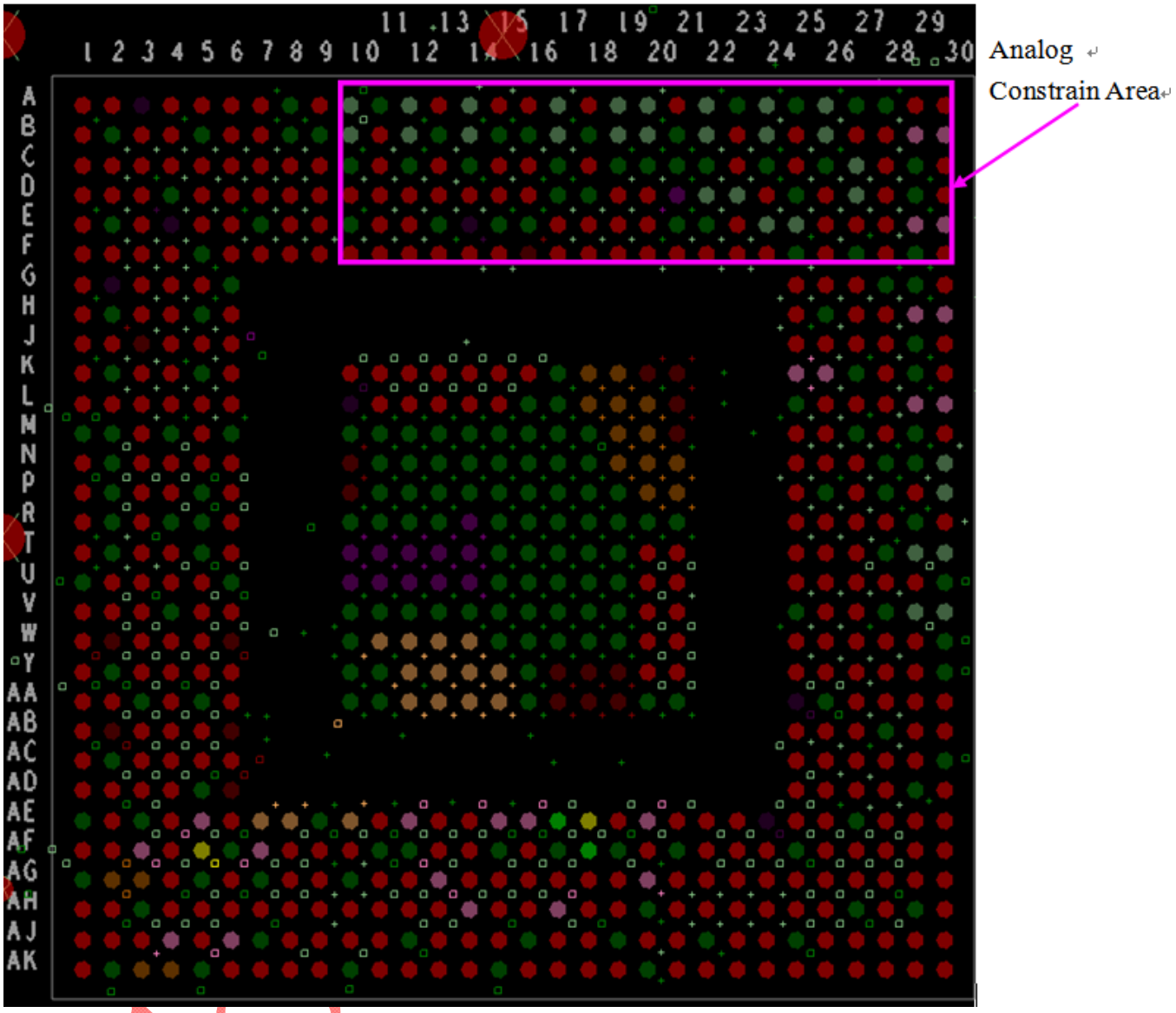


12. Routing Vref as short as possible with minimum 30mil trace width and at least 30mil space from other signal. Routing trace should keep minimum 5 times trace width of next trace between high speed signals to reduce crosstalk.

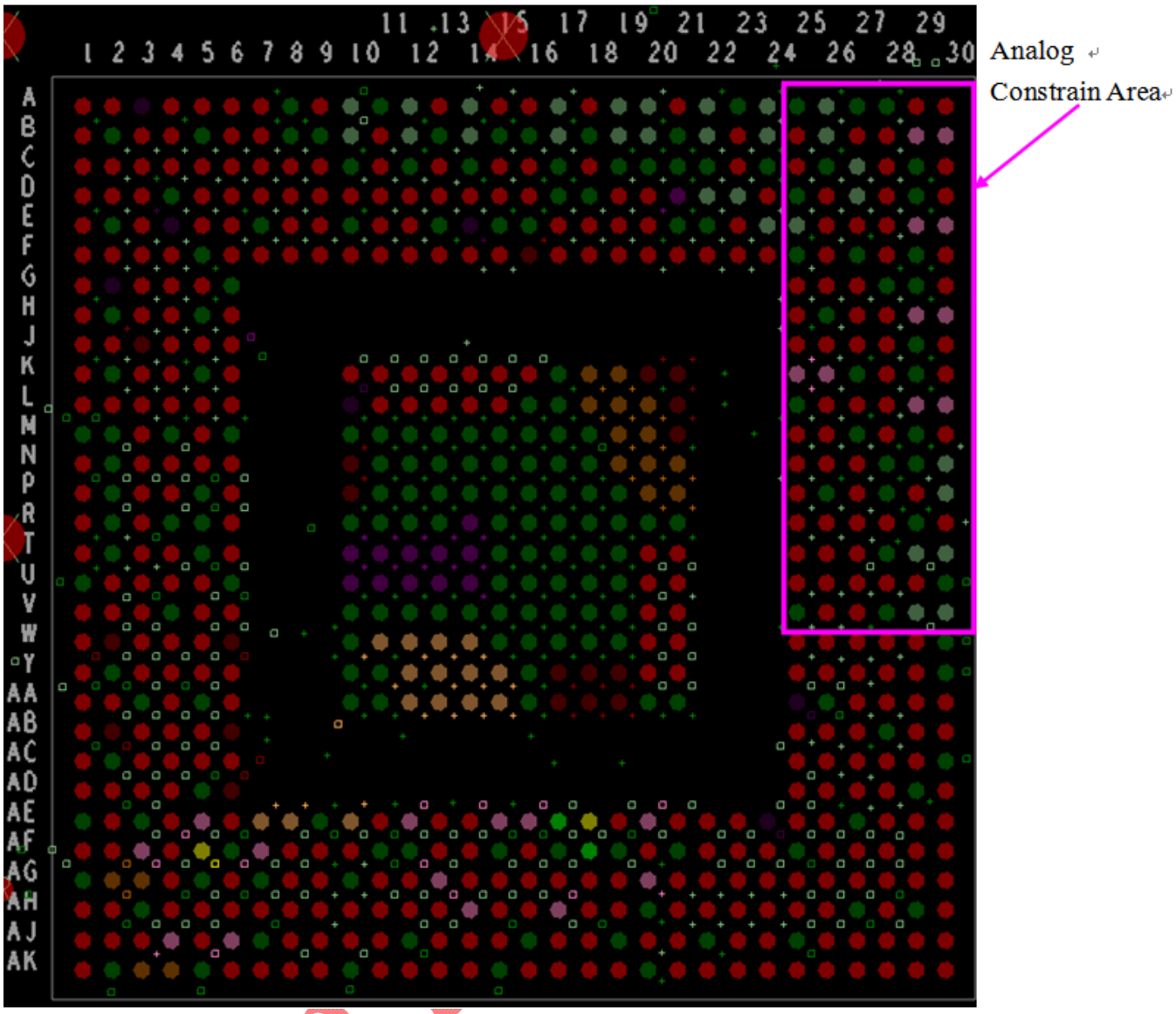
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1.12. Analog area Layout Guideline

Avoid any digital signal cross the area. The constraint area for VORTEX86DX3 is as follows:



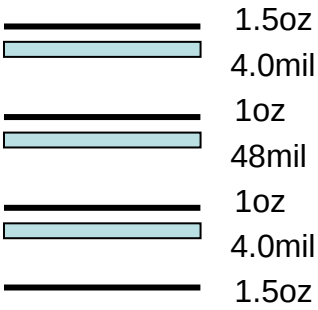
The constraint area for VORTEX86DX3 is as follows:



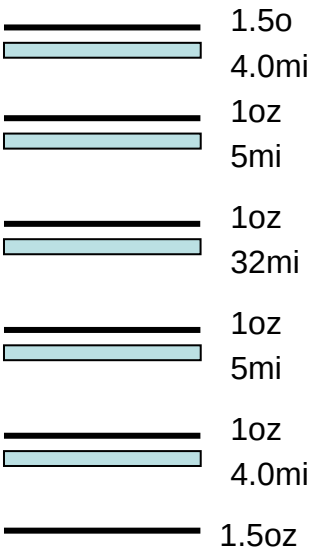
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1.13. PCB Stack up

The recommendation PCB stack up is as follows for 4 layers PCB:



The recommendation PCB stack up is as follows for 6 layers PCB:



1.14. Bypass Capacitor scheme

The role of the bypass capacitor is to decouple the power supply bus from the IC. Bypass a power bus from the device internal noise requires capacitor with very low inductance. The objective is to eliminate the effects of the power bus inductance and resistance so that transient currents flowing across the power bus do not cause excessive noise at the power and ground pins of the IC. Therefore, the bypass capacitor should have low effective series resistance (ESR) and series inductance (ESL) while having a large enough capacitance value to supply current to the IC during switching.

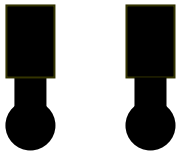
The recommendation bypass scheme for the large surge current is as follows:

1. It is recommended that at least a via per power pin for SDRAM memory interface
2. It is recommended that multiple vias per bypass capacitor to further reduce inductance.
3. It is recommended that X7R or X5R type and 0402 size capacitor for bypass.

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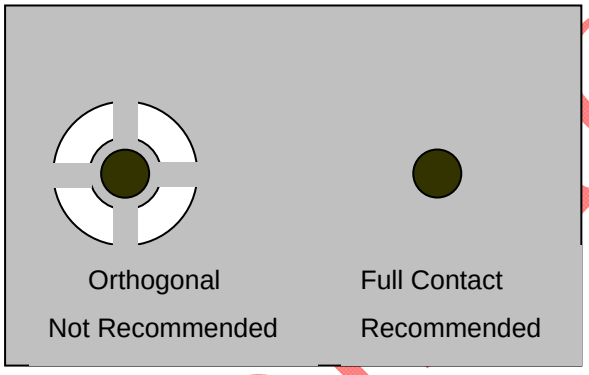
4. It is recommended that a 47uf bulk capacitor for the interface.

5. The land pattern of bypass capacitor is as follows:



1.15. Suggestion Via for DDR3 and VORTEX86DX3

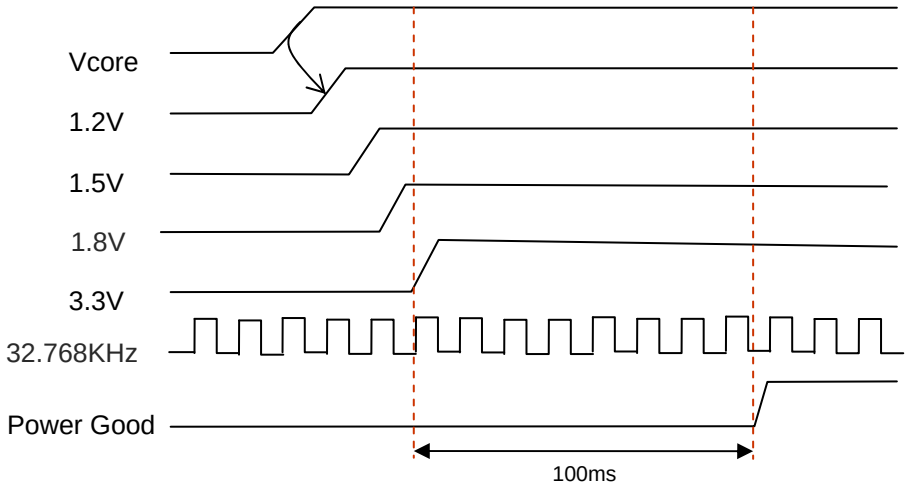
In order to minimum the series inductance, ground and power via should be maximized as possible. We are suggestion on DDR deice that via is at least 18 mil pads with 10mil drilling size. We are also suggestion on VORTEX86DX3 device that via is at least 20 mil pads with 10mil drilling size. The via pattern connect to power/ground plan is as follows:



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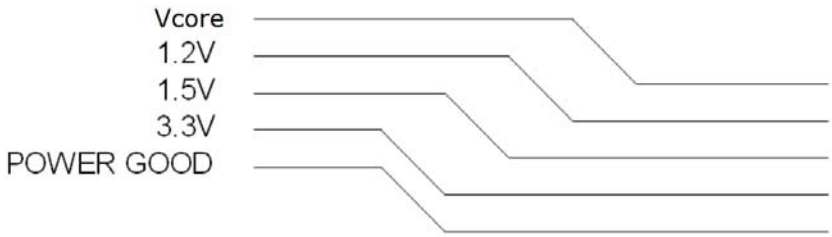
1.16. POWER ON SEQUENCE

The recommend power up sequence is as follows:



1.17. POWER OFF SEQUENCE WAVEFORM

The recommended power down sequence is as follows:



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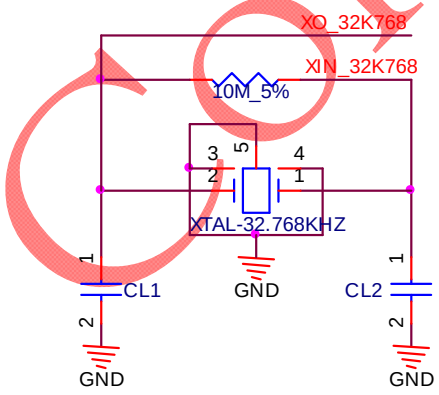
1.18. Crystal recommend

The recommend specification of 32.768KHz crystal is as follows:

PACKAGE : I308 3.0 x 8.0 mm
ITTI PART NO. : 1300075-32.768KHZ
CUSTOMER PART NO. :
ELECTRICAL CHARACTERISTICS :

Parameter	Symbol	Condition	Specification	Unit
Nominal Frequency	Fo	At 25°C	32.768	KHz
Frequency Tolerance	ΔF/Fo	At 25°C	±20	ppm
Temperature Coefficient	Tc		-0.035±0.01	ppm/°C²
Operating Temp. Range	Topr		-10 to +60	°C
Storage Temp. Range	Tstg		-40 to +85	°C
Load Capacitance	CL		12.5	pF
Equivalent Series Resistance	R1		35 max.	kΩ
Drive Level	DL		1 max.	μW
Aging	Fa	At 25°C, Per Year	±5.0 max.	ppm

The recommend circuit is as follows:



The parasitical capacitance of 5mil PCB trace width on outer layer is about 1.1835pF/cm for 3.8mil P.P reference to shield layer.

The total capacitance of CL1+x*1.1835+3=CL1_total

Where x is trace length crystal pin to Xout pin of VORTEX86DX3.

The total capacitance of CL2+y*1.1835+3=CL2_total

Where y is trace length crystal pin to Xin pin of VORTEX86DX3.

CL=CL1_total*CL2_total/(CL1_total+CL2_total)

CL is the load capacitance of crystal.

The recommend specification of 14.318MHz crystal is as follows:

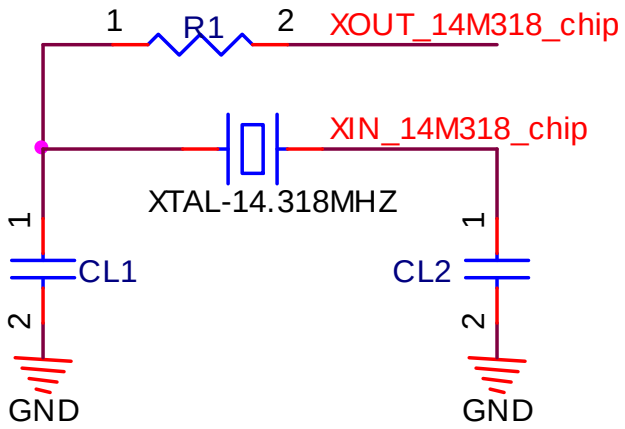
PACKAGE : L50 (5.0 x 3.2 x 1.3) mm

ITTI PART NO. : 1300712-14.31818MHz

ELECTRICAL CHARACTERISTICS :

Parameter	Symbol	Condition	Specification	Unit
1. Nominal Frequency	Fo	At 25°C	14.31818	MHz
2. Oscillation Mode	Fundamental			
3. Frequency Tolerance	ΔF/Fo	At 25°C	± 20	ppm
4. Tolerance Stability Over Temp. Range	Tc		± 20	ppm
5. Operating Temp. Range	Topr		-20 to +80	°C
6. Storage Temp. Range	Tstg		-40 to +85	°C
7. Load Capacitance	CL		12	pF
8. Equivalent Series Resistance	R1		60 max.	Ω
9. Drive Level	DL		300 max.	μW
10. Aging	Fa	At 25°C, Per Year	± 5 max.	ppm

The recommend circuit of 14.318MHz crystal is as follows:



The R1 is damping resistor to control the drive level under the crystal specification.

R1 of 470ohm is recommended for the 14.318MHz crystal.

The total capacitance of CL1+x*1.1835=CL1_total

Where x is trace length crystal pin to R1 pin of damping resistor

The total capacitance of CL2+y*1.1835+3=CL2_total

Where y is trace length crystal pin to Xin pin of VORTEX86DX3.

CL=CL1_total*CL2_total/(CL1_total+CL2_total)

CL is the load capacitance of crystal.

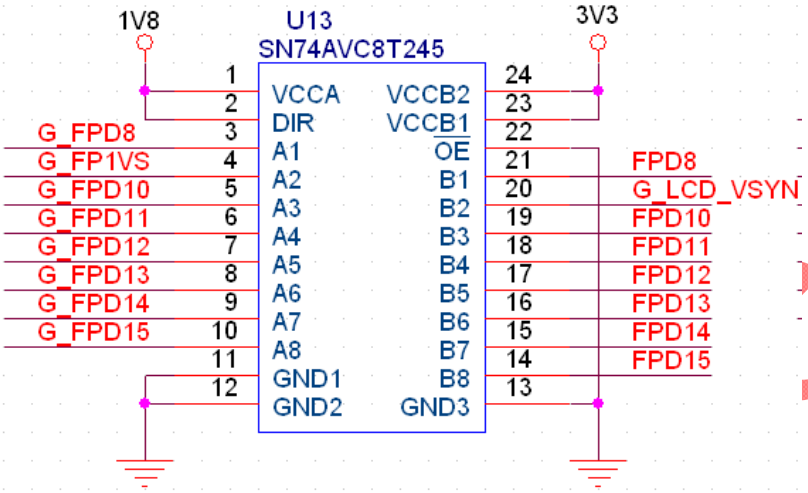
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1.19. DVO layout guideline

According to the below-mentioned guideline, some constraints were appended as follows:

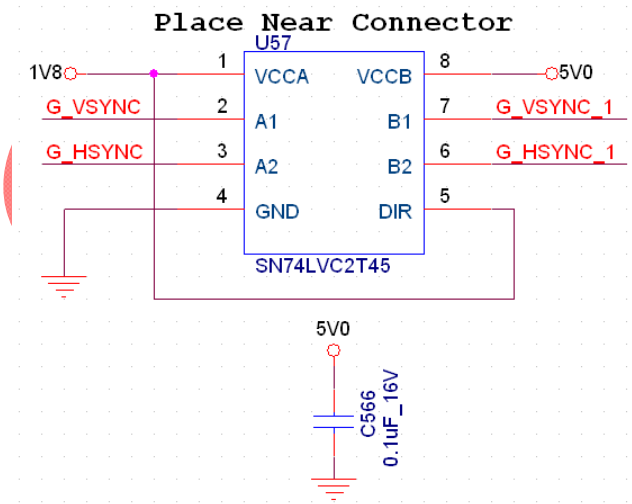
1. Control the impedance of trace between 55 and 60 ohm.
2. DVO CLK with 3W space separate other signal.
3. Please damping resistor near chip.
4. DVO signal space with 3W space separate other non DVO signal.
5. Skew of DVO0~DVO23 signal control within 500mils.
6. If DVO signal has STRAP feature unusable with BUS HOLD function level shift chip.

The recommend circuit of level shift is as follow:

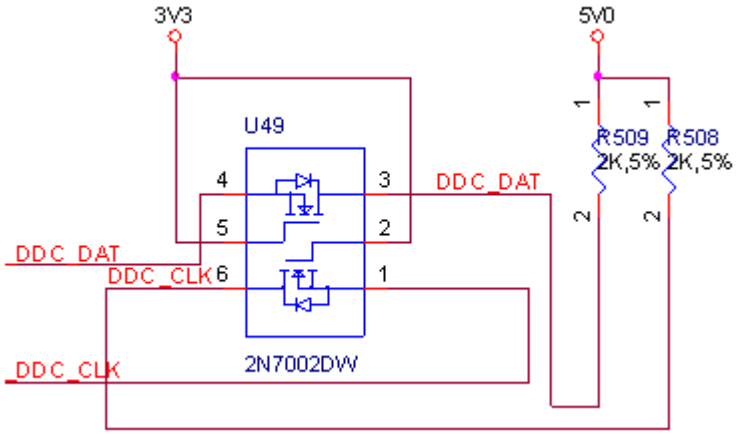


1.20. The reference circuit for VSyn and Hsyn of VGA

To drive traditional CRT monitor by our VGA port. We recommend adding level shift on the output of Hsyn, Vsyn and DDC channel.



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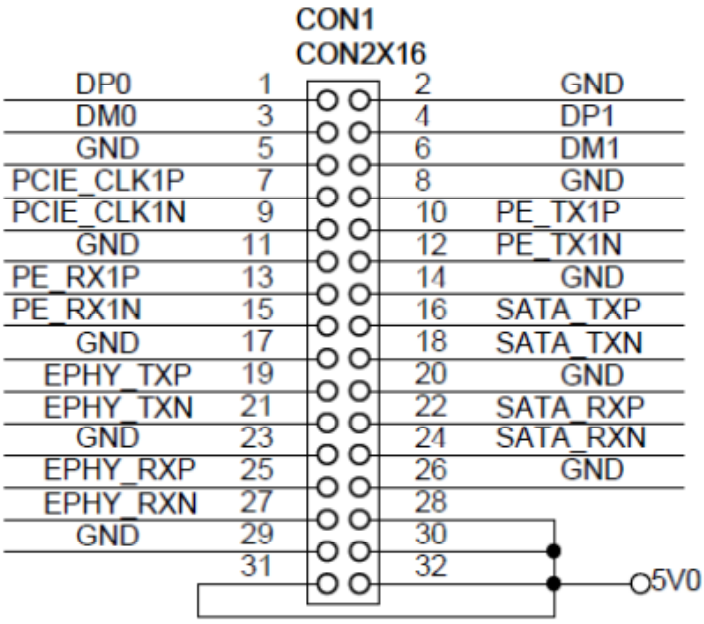
1.21. Pins connection of PCIe and SATA interface if no need

1. For PCIe power interface, connect power pin(pin D24,B23,B25,B26 and B27,) to 1.2V,connect power pin(pin C25) to 3.3V.
   For PCIe control and clock interface, to leave unconnected for control pins of A24,B24,A22,B22,A26,B26,C27, and D27.To leave unconnected for clock pins of E25,E24,A19,B19,A17 and B17.
2. For SATA power interface, connect power pin(pin W29,U29,R30) to 1.2V,connect power pin(pin P29) to 3.3V,
   For SATA control and clock interface, to leave unconnected for control pins of T30,T29,V30 and V29. To leave unconnected for clock pins of P30,N30,A20 and B20.

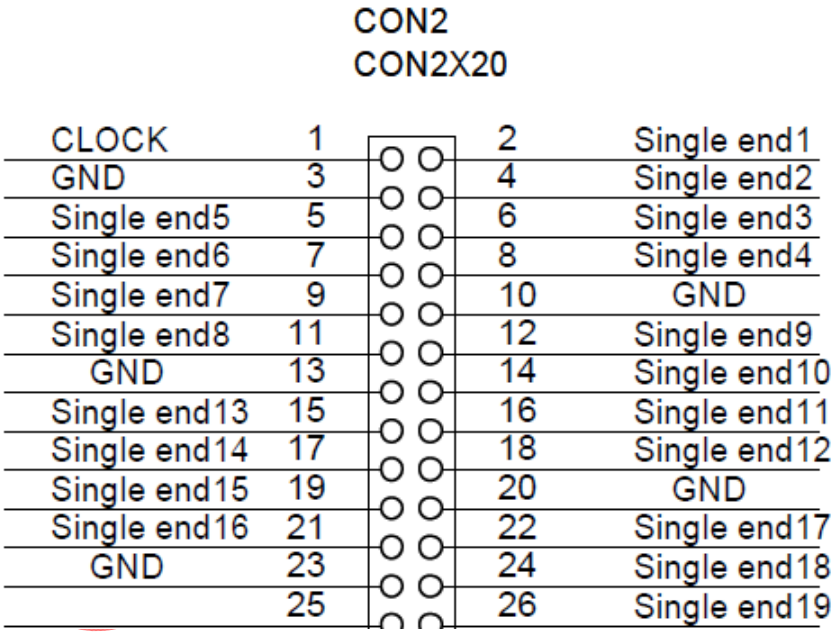
1.22. Pins connection define for high speed signal connect to other board by board to board connector

1. For high speed signal connector pin define, like as USB differential pair , PCIe clock differential pair and control differential pair, SATA control differential pair , Ethernet control differential pair all these high speed signal, as follows example connector is high speed signal pin define recommend to reduce crosstalk:

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2. For high speed single end signal connector pin define , like as clock signal, next pin should be defining to GND signal for shielding , otherwise , define 1 net of GND signal between each 4 net of single end signal to reduce crosstalk ,example connector pin define as follows:

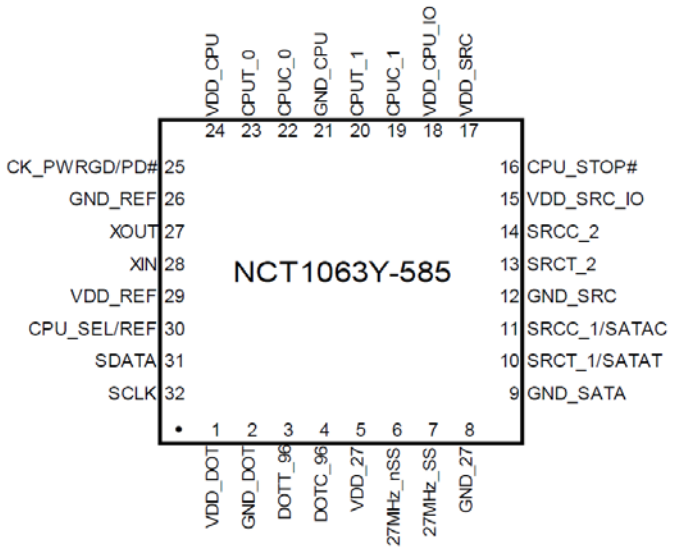


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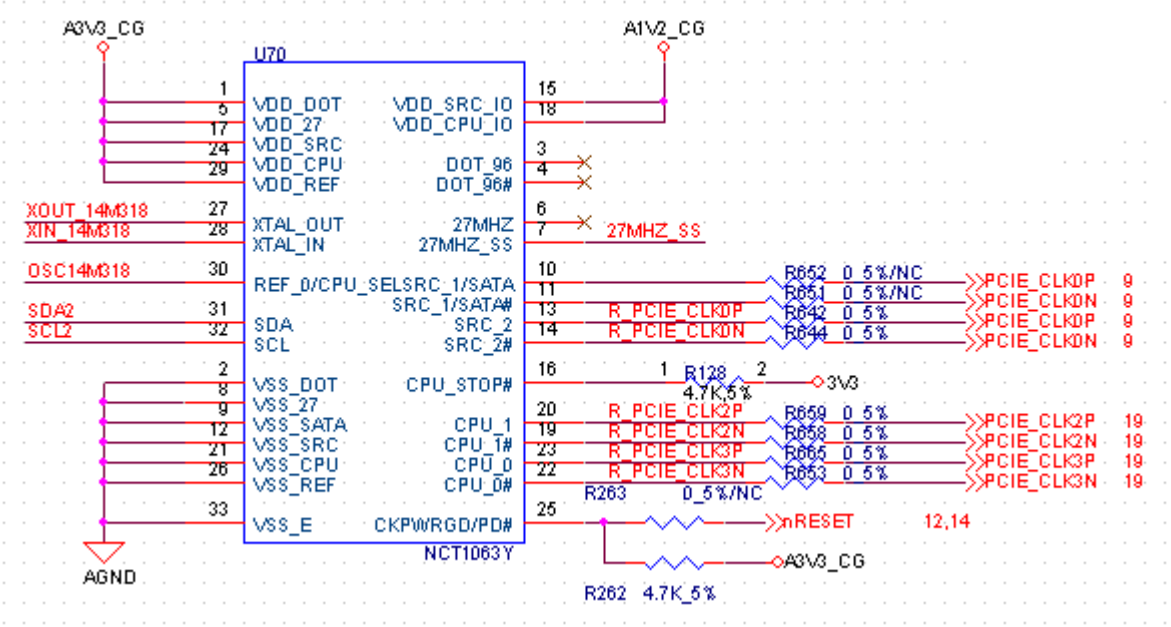
1.23. PCIe External clock guideline

1. The external clock generator part number is **NCT1063Y-585** that we recommend. As follow pin out is specification of external clock generator:

PIN CONFIGURATION



2. Differential pairs of Clock generator, pin of 10/11 connect to VORTEX86DX3 pin of E25/E24 for PCIe internal PLL, and pin of 13/14 and 20/19 connect to PCIe device's differential input clock pair. The two trace width is 4.5 mil and space 5mil to get near 100ohm differential impedance, and keep minimum 30 mils space for other traces to reduce crosstalk as possible. The trace length of two differential signals must be match within 10mil. As follow circuit is specification of external clock generator that we recommend:

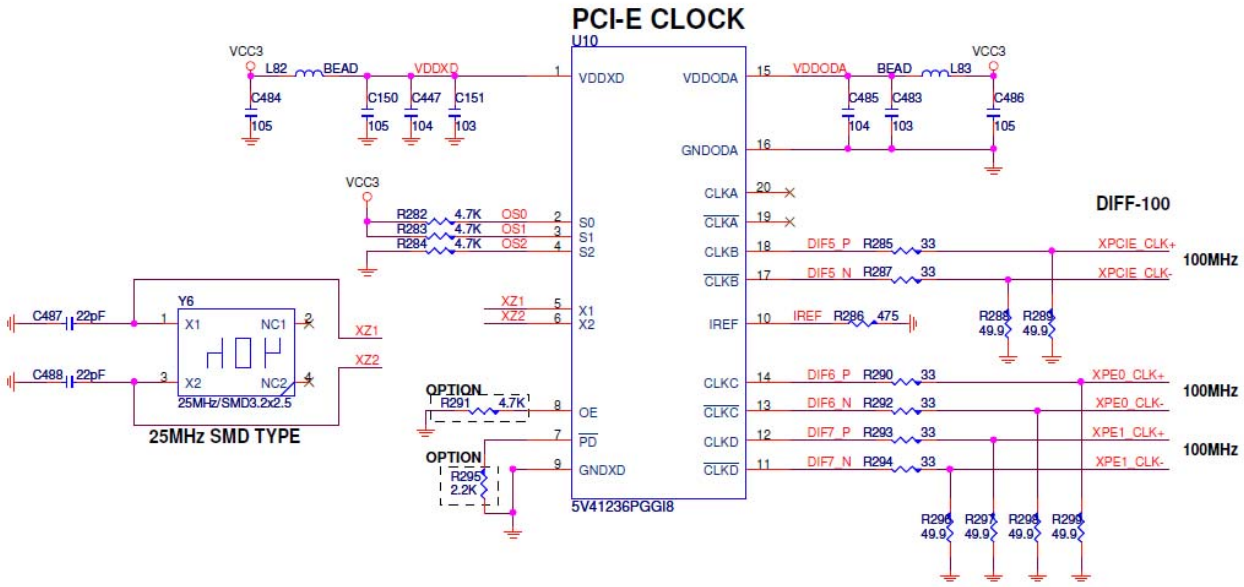


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3. Clock generator recommend list.

Vendor	Part number
NUVOTON	NCT1063Y-585
IDT	ICS843751
SILEGO	SLG8SP585
IDT	IDT5V41236

For new design, DMP suggest to use IDT5V41236, please see the reference schematic as below.



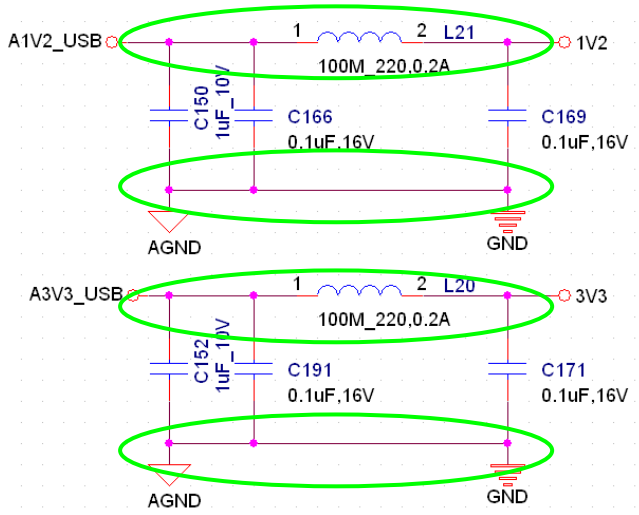
1.24. USB Port Power Source Notice

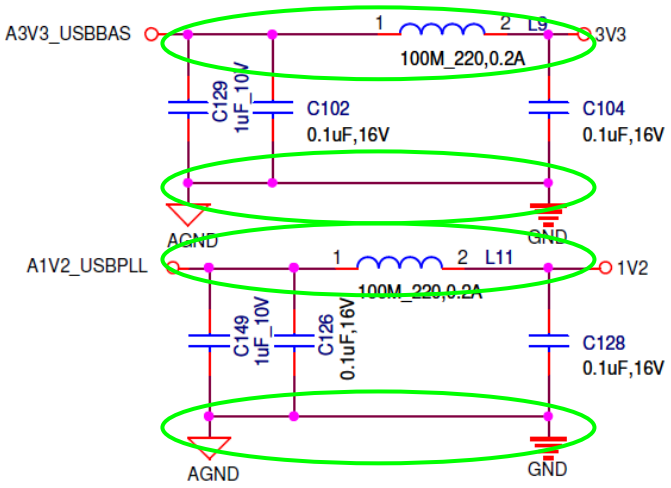
To avoid USB device power low drop cause system power jitter noise, tuning up to 5.2V for USB port power source is recommend.

1.25. USB power layout Guideline

A clear power supply is importance for USB. DMP recommend a filter technology to reduce jitter as follows:

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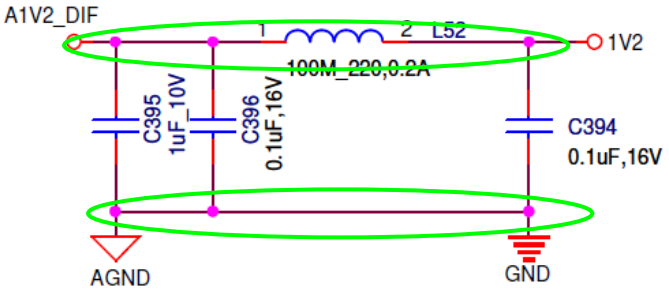


Minimum 30mils width of trace is preferable.

Rely on VORTEX86DX3 chip side the second layer must be the ground layer.

1.26. Clock PLL power layout Guideline

A clear power supply is importance for Clock PLL. DMP recommend a filter technology to reduce jitter as follows:

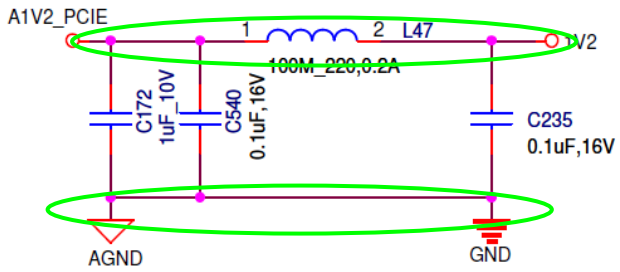


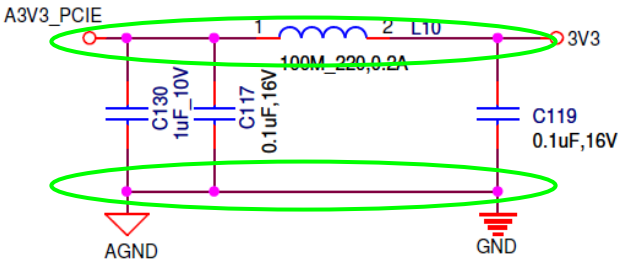
Minimum 30mils width of trace is preferable.

Rely on VORTEX86DX3 chip side the second layer must be the ground layer.

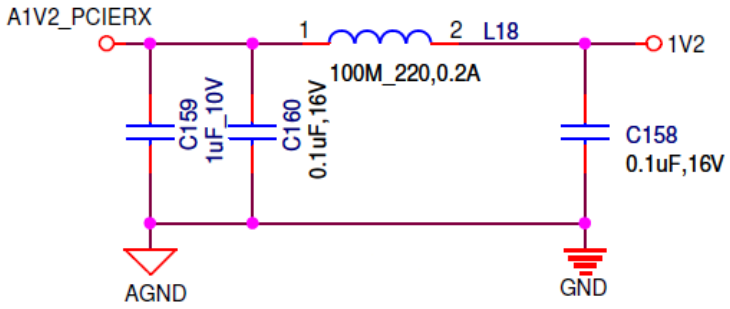
1.27. PCIe power layout Guideline

A clear power supply is importance for PCIe. DMP recommend a filter technology to reduce jitter as follows:





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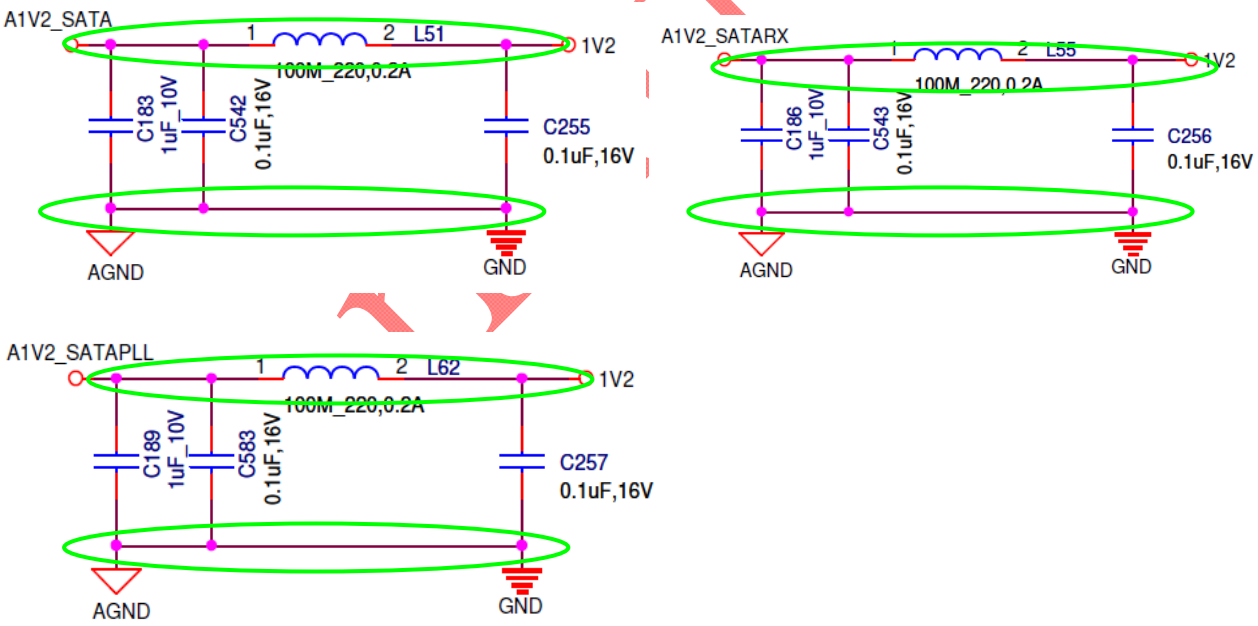


Minimum 30mils width of trace is preferable.

Rely on VORTEX86DX3 chip side the second layer must be the ground layer.

1.28. SATA power layout Guideline

A clear power supply is importance for SATA. DMP recommend a filter technology to reduce jitter as follows:



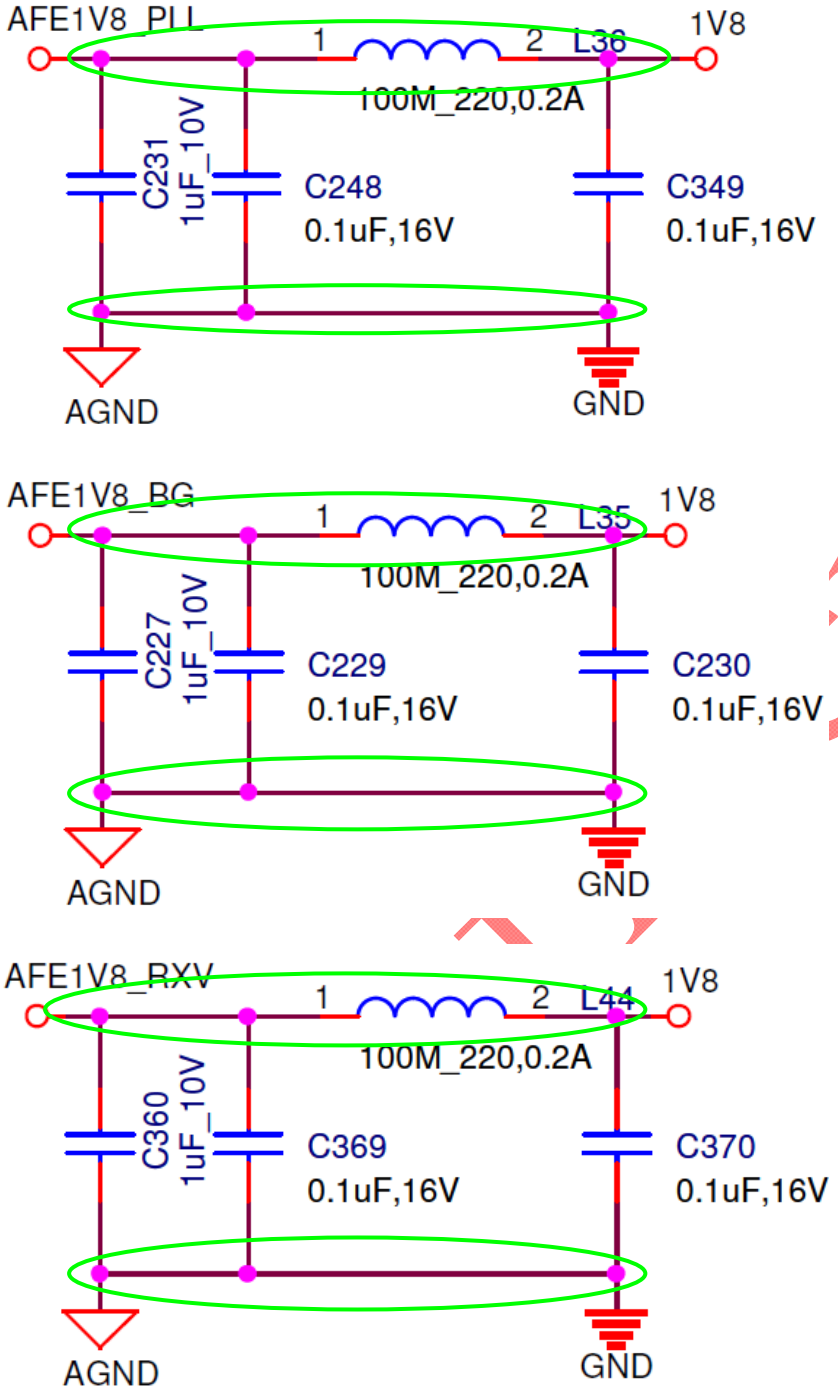
Minimum 30mils width of trace is preferable.

Rely on VORTEX86DX3 chip side the second layer must be the ground layer.

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1.29. Ethernet power layout Guideline

A clear power supply is importance for Ethernet. DMP recommend a filter technology to reduce jitter as follows:



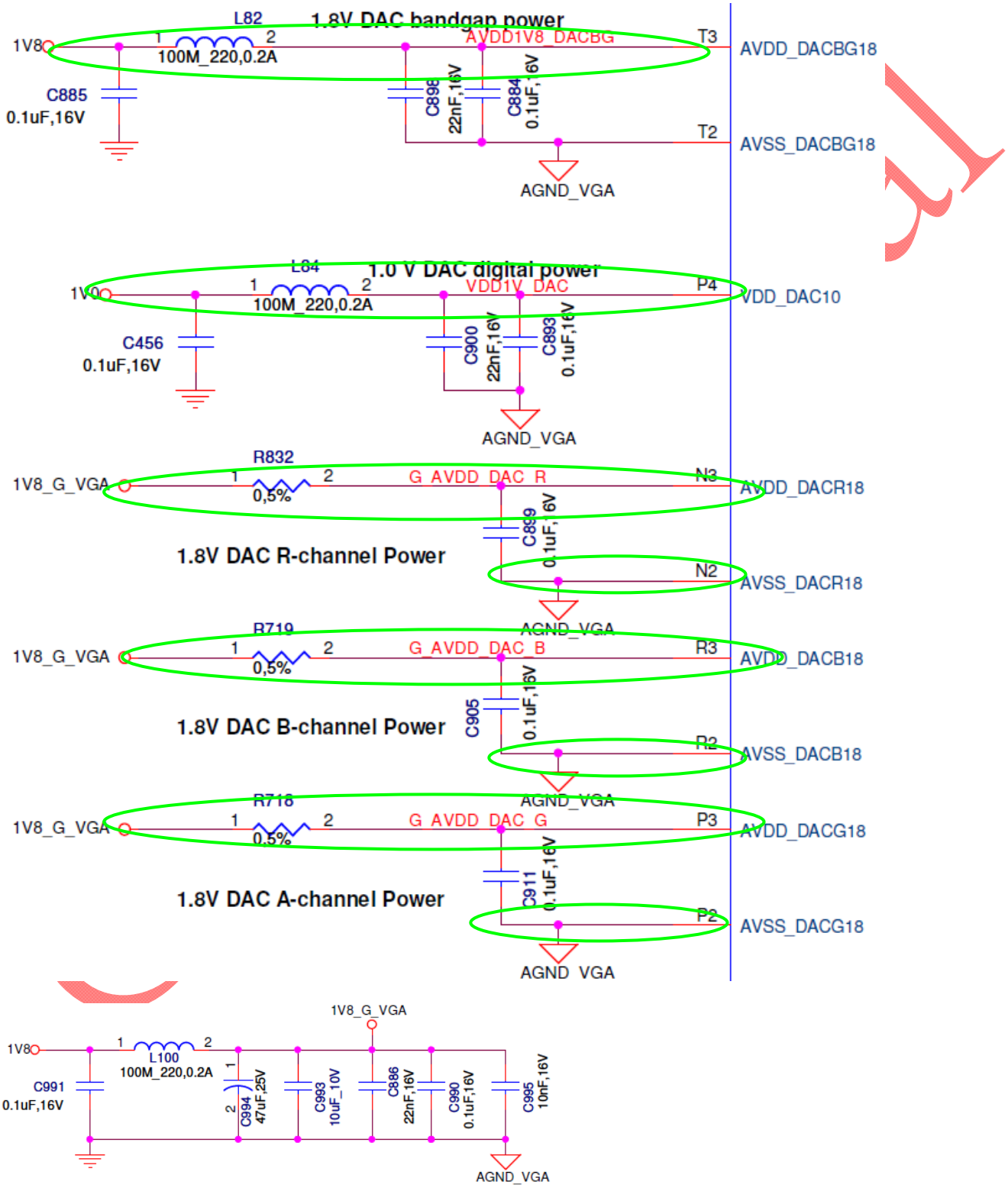
Minimum 30mils width of trace is preferable.

Rely on VORTEX86DX3 chip side the second layer must be the ground layer.

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1.30. DVO power layout Guideline

A clear power supply is importance for DVO. DMP recommend a filter technology to reduce jitter as follows:

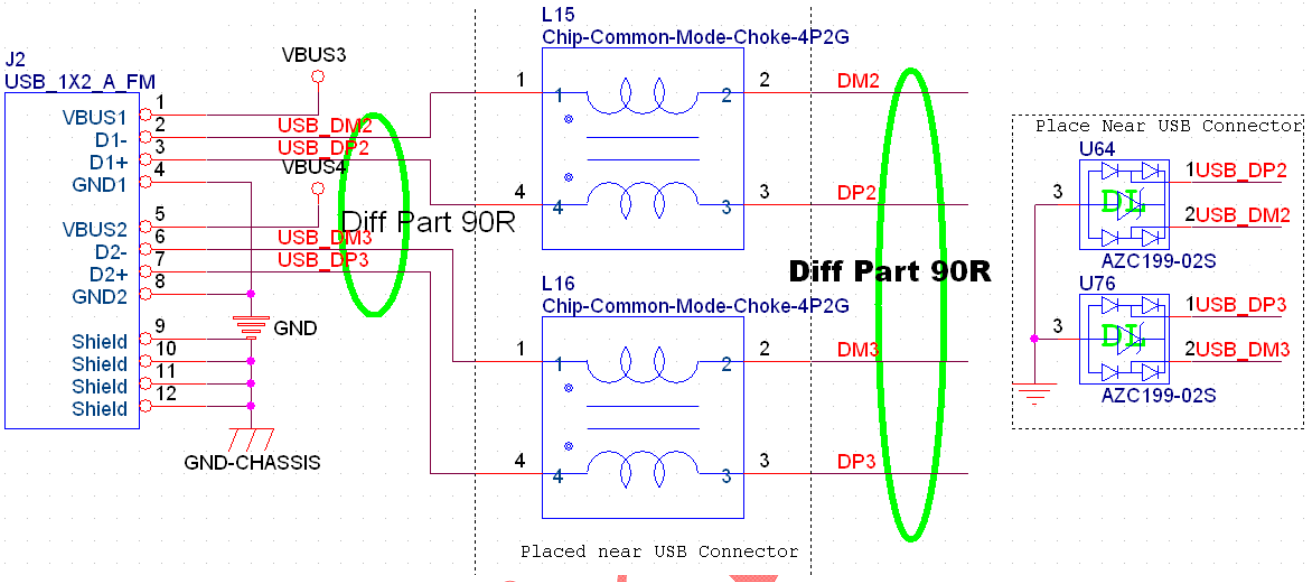


Minimum 30mils width of trace is preferable.

Rely on VORTEX86DX3 chip side the second layer must be the ground layer.

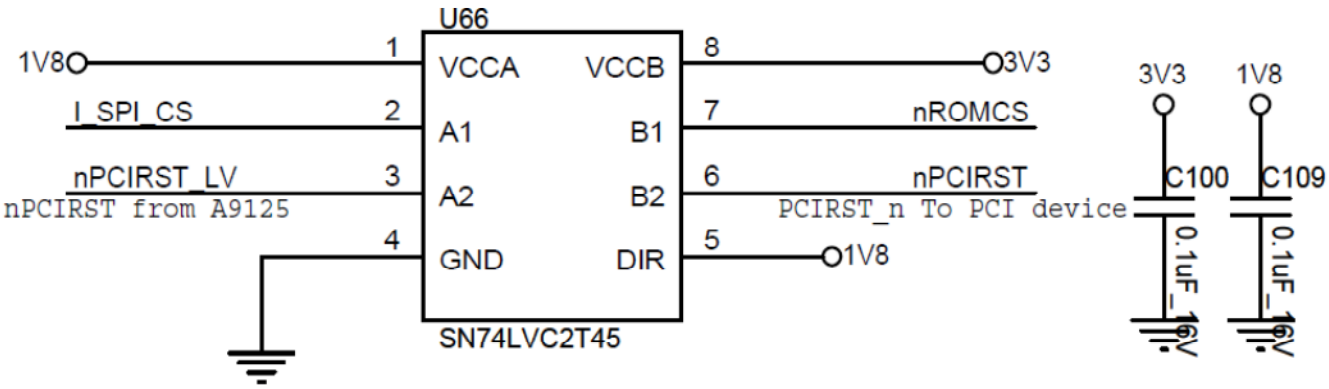
1.31. USB ESD layout Guideline

To avoid high voltage pulse cause USB fail issue , place TVS protect component for ESD solution, and TVS protect circuit have to use 1 high-speed I/O lines specification of TVS for USBD+ and USBD- differential pairs , TVS component place near by USB Connector as following:



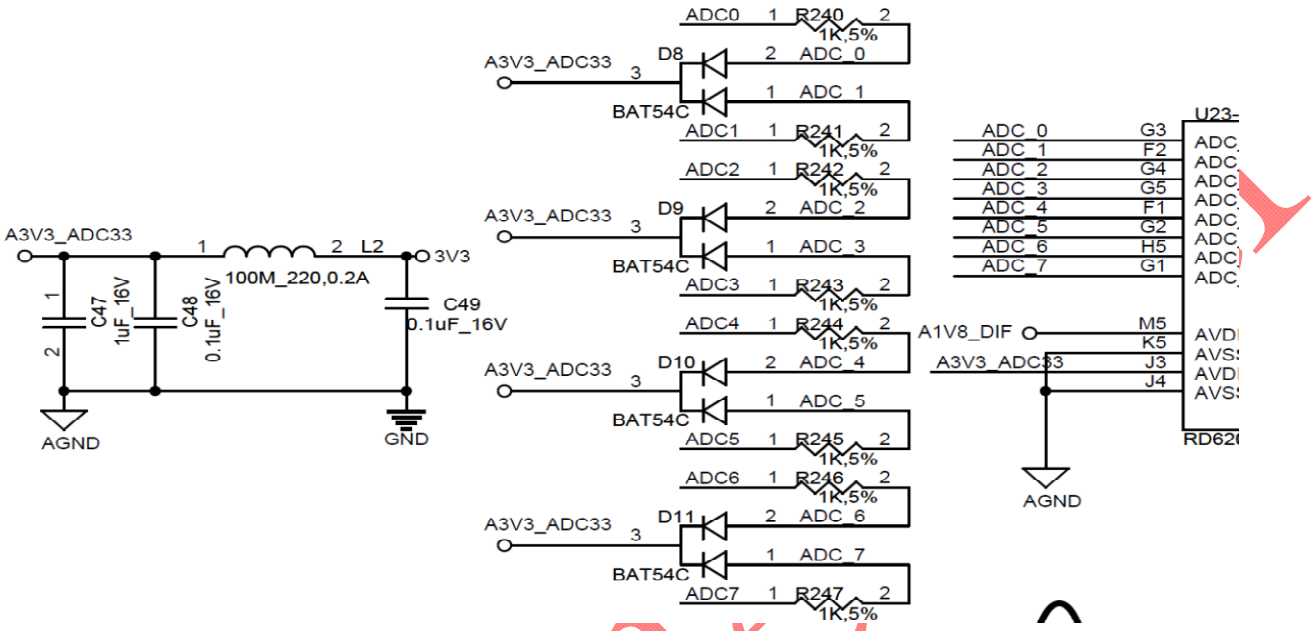
1.32. PCI Reset signal recommend circuit

PCIRST# is output signal from A9125, this pin is used to reset PCI device, when it asserted low, all of PCI device will be reset, it is 1.8V PCI reset pin, to avoid PCI device reset input pin is asserted error, have to add as following circuit:



1.33. ADC Port Input Circuit Protection Notice

To avoid ADC input high pulse voltage made data read error from register , have to add as following circuit to protect ADC input normal function:



1.34. VGA layout Guideline

Place termination resistor of 150 ohm near CRT connector and VORTEX86DX3 side and the EMI filters near CRT connector for signal G_IOR, G_IOG and G_IOB. Control the trace impedance of G_IOR, G_IOG and G_IOB to 50ohm. Place protector of ESD near CRT connector. Place the resistor of ISET near VORTEX86DX3 at least with trace width of 20mil. The circuit of suggestion is as follows:

